1 . **Position Information**	
Job Title:	**Consultancy to Develop a Traditional Marriage Certificate Software for the Ministry of Internal Affairs**
Contract Type:	
Reports to:	**Director of IT, Ministry of Internal Affairs**
Duty Station:	**Monrovia**
Duration of Contract:	**34 Days over 3 Months**

II. Background

Decentralization has been a core reform objective of the Liberian Government over the past several years. The government's interest in this reform is premised on Liberia's perennial problem of over-centralization of public administration, which has long undermined popular participation in national decision-making and limited access to social services and economic opportunities for most Liberians. Since the inception of the decentralization reform, the government has demonstrated immense political will by matching its commitments with concrete actions.

In 2012, a National Policy on Decentralization and Local Governance was launched to establish a framework for effective and efficient service delivery at the sub-national level, promote participatory decision-making to foster peacebuilding and national reconciliation, and strengthen local planning, monitoring, and management capacities. Building on this momentum, the Government of Liberia, with support from UNDP and other development partners, established the Liberia Decentralization Support Program (LDSP) as a vehicle for resource mobilization and coordinated program implementation to make decentralization a tangible reality.

Within this broader framework, the development of a Traditional Marriage Certificate Software aligns with the government's ongoing efforts to digitize and decentralize public services. Traditional marriage, which remains a deeply rooted social and cultural institution in Liberia, has historically faced administrative and documentation challenges, including inconsistent recordkeeping, limited accessibility of marriage certificates in rural areas, and difficulties in verifying marital status. By developing a digital platform to register, authenticate, and issue traditional marriage certificates, the government seeks to enhance service accessibility, transparency, and data management at local levels.

Furthermore, this initiative supports the decentralization agenda by empowering local authorities particularly within county and district administrations to efficiently manage civil registration services while ensuring that records are securely stored and easily retrievable. The software will also help manage traditional marriage records into the national civil registry system, thereby promoting gender equality, protecting inheritance and property rights, and contributing to improved governance and planning through reliable data collection.

Accordingly, this consultancy has been designed to provide the technical expertise required to guide the development and successful deployment of the Traditional Marriage Certificate Software. The consultant will lead the end-to-end process from system design and development to testing, validation, and rollout ensuring that all functional, technical, and user requirements are adequately addressed. This exercise will not only demonstrate the operational effectiveness of the digital platform but also generate practical lessons to inform future digitalization initiatives under the decentralization program. Moreover, it will strengthen institutional capacity, promote data-driven management of traditional marriage records, and contribute to establishing a sustainable digital framework for local governance in Liberia.

III. Scope of Work

Under the overall supervision of the Inclusive Governance Team Leader and in direct coordination with the LDSP Interim Programme Coordinator, the consultant will undertake the following tasks:

System Design:

Authentication and Authorization

- The system must provide an authentication mechanism for ensuring that only registered users have access to the database system.
- Once a user has been authenticated, the system must apply access control mechanisms to ensure that a user's access to system functions and data resources corresponds with their assigned level of authority or access rights and privileges.
- The system must also provide a means of separating roles and duties. For example, a role-based access control might be used to limit access to data resources at folder and data levels.
- The system must provide the ability to logout a user's account after three failed logon attempts as well as provide a means of recording and retaining both successful and failed logon attempts.
- The system should also provide the ability to enforce minimum password length and strength as well as password expiration rules.

Database Management Tools

- The system must provide the administrative tools necessary for managing the database, including the creation and assignment of user rights and privileges; the deletion of users and modification of access rights; scheduling of events; reports generation; backup and recovery; database optimization; and general database security management.
- In terms of backup mechanism, the system should also support dynamic backup.

Automatic Alert Trigger

- The system must have the ability to create an alert to any life-event record, which is automatically activated if certain conditions are met.

Interface

- The system must provide a Graphical User Interface (GUI) for data entry and associated activities. The interface must be user friendly, clear, predictable, consistent and efficient.

Database Security

- Due to the sensitive nature of the data elements that are intended to be held in the database, the system must provide an encryption mechanism for securing the entire database (data at rest) as well as ensuring that transmitted data between two end points (externally and internally) are not compromised.
- As encryption affects data size and impacts on database performance, the system architecture should be designed to minimize any performance degradation.

Audit Trail and Log

Commented [ZMG1]: How will the software be hosted? Will the software be hosted in the cloud or on premises?

Will the software have the following features?

Offline & Mobile Support.
Data Quality & Duplicate Checks.
Integration & APIs.
Performance & Scalability.(Capable of supporting concurrent users per agreed SLA; scalable for national deployment.
Response times: <2s for simple queries under normal load; acceptable benchmarks in TSD.)

Commented [L2R1]: The Software will be hosted by the LIBTELCO through negotiation between head of entities

The Software will need to be integrated with other systems like the E-Liberia portal and the LRA platforms etc. including Data quality, APIs. Due to the remote counties, yes we need the software to be offline and mobile support

- The system must provide a secure and automated audit capability for recording activities in the system as well as changes made to data records. All suspicious activities must be logged and reported to the Administrator.
- The proposed system must record the following audit trail activities:
  - ✓ Nature of the event (for example, access to records, insertion and deletion)
  - ✓ Identity of the user or system component
  - ✓ Point of origin of the event including Date and time
  - ✓ Identity of data or system resources affected
- The system must record and keep information about unsuccessful attempts to access data and system resources. The log file must be secured, and only authorized people should have access to the file.
- The system must provide a means of ensuring that the audit trails and log files cannot be altered or deleted.

Production and Printing

- The system must provide the ability to produce, view and print, in so doing, the system must be able to produce individualized profiles by allowing automatic extraction of relevant data from the database.

Reports Generation

- The proposed system must provide the ability to generate, print, and/or save different types of reports including predefined routine reports as well as ad-hoc reports.
- The reporting capability must include the generation of reports using multiple categories as well as provide automatic calculation features. The system must also have the ability to generate, view, print, and/or save audit reports, including exception reports.

Data Validation

- The proposed system must provide the ability to validate data inputs. Data validation functions are required to ensure data entry completion and to prevent the entry of invalid/incorrect information into the database. As input validation is crucial to the overall functioning of the database, the proposed system must ensure that illegal or invalid values are not passed to the database.
- The system must provide a means of ensuring that the user has completed all mandatory data entry fields before enabling the registration of a new record.
- The system must have the ability to display warning/alert messages as well as provide the ability for soft and hard edits. For example, if the date entered is in the future, an instructive pop-up warning/message should inform the user of the data entry error as well as the reasons for the error (context sensitive help).

Data Input Format Correction

- The system must provide the ability to change certain data inputs to conform to the desired format. For example, if the date does not conform to the desired format, the system should automatically change it to conform rather than reject the input. The system must have the ability to capitalize (auto covert) all proper nouns.For example, if the first letter of a name is not capitalized, the system should automatically capitalize it.
- The proposed system must ensure that the date text fields contain a pop-up calendar.

Search and Retrieve

- The proposed system must provide the functionality for searching and retrieving any

Commented [ZMG3]: Search, Verification & Reporting.

Advanced search: by names, dates, certificate number, location, party ID, or custom fields.
Quick verification feature: generate scannable QR code or unique certificate ID for public/third-party verification.
Pre-built reports (monthly registrations, demographics, duplicate detection) and ad-hoc report builder.
Export capabilities: CSV, Excel, JSON, and XML.

Commented [L4R3]:

Commented [L5R3]: These are all features that will be needed potential duplicate record before enabling the creation of a new life event record in the database. If a duplication record is found, the system must provide a means of handling the duplicate record.

- The system must provide the ability to search for records of employees using either a single criterion or multiple criteria. The system must also provide support for wild card searches.
- The system must ensure that the search functionality is not case sensitive.
- The system must provide the functionality for the user to sort search results in either ascending or descending order.

Deletion of Invalid Records

- As there may be cases where completed records of life events are no longer valid for one reason or another (for example, an erroneously or fraudulently created record), the system must provide a means of voiding such entries. However, the system must ensure that invalid records can only be voided and not deleted from the database.
- The system must also provide the ability to permanently archive voided records, as well as a means of retrieving and viewing these records. It must also provide a way of ensuring that only authorized high-level users have access to all voided or cancelled life events records.

Keystrokes Reduction and Data Entry

- The system must provide mechanisms to enable data entry efficiency and keystroke reduction wherever practicable. For example, an auto-populate functionality can be used to populate certain text fields based on input selection from a drop-down menu.
- The system must provide the ability to dynamically suppress or disable text fields based on the data entered in the previous fields.

Future Technological Integration

- The system should be able to interface with other local automated systems.
- The system must have the ability to securely interconnect with these systems via a web-based interface. The web-based interface should support the use of Secure Socket Layer (SSL) for data transmission.
- The proposed system must ensure that incoming data input are validated, evaluated for expected size, format and type before acceptance. Because of the sensitive nature of the information being transmitted, the system must have the ability to reject unencrypted data.

System Validation

- Develop and Implement a Validation Plan for the Traditional Marriage Certificate Software;
- Conduct Functional Testing of all system modules and functionalities to ensure that they meet user requirements.
- Perform User Acceptance Testing with the Ministry of Internal Affairs;
- Verify Data Integrity and Security to assess the software protection mechanism and user access controls;
- Evaluate System Performance and Scalability to assess system speed, responsiveness, and load capacity under different usage conditions;
- Validate Integration with existing systems to ensure compatibility with the GOL ICT standards;
- Prepare Validation Report

Training of ICT/CSC Focal Persons

- Prepare a detailed training plan specifying learning objectives, training content, delivery approach, duration, and participant selection criteria;
- Develop tailored training modules focusing on system operation, user management, data entry, verification, certificate generation, and reporting functions;
- Produce user-friendly materials such as step-by-step guides, manuals, and quick reference sheets to support CSC focal persons in day-to-day operations;
- Provide post-training support and mentorship;
- Submit training report.

System Maintenance and Technical Support-

- Implement a security architecture to protect data integrity and user information and ensure operational and maintenance (O&M), provide ongoing capacity support to ensure smooth operation of the platform, and offer technical support and maintenance services for 6 months post deployment.

IV. Deliverables	Days
System Design Authentication and Authorization: The proposed system must provide an authentication mechanism for ensuring that only registered users have access to the database system. Once a user has been authenticated, the system must apply access control mechanisms to ensure that a user's access to system functions and data resources corresponds with their assigned level of authority or access rights and privileges. The system must also provide a means of separating roles and duties. For example, a role- based access control might be used to limit access to data resources at folder and data levels. The system must provide the ability to logout a user's account after three failed logon attempts as well as provide a means of recording and retaining both successful and failed logon attempts. The proposed system should also provide the ability to enforce minimum password length and strength as well as password expiration rules. Database Management Tools: The system must provide the administrative tools necessary for managing the database, including the creation and assignment of user rights and privileges; the deletion of users and modification of access rights; scheduling of events; reports generation; backup and recovery; database optimization; and general database security management. In terms of backup mechanism, the system should also support dynamic backup. Automatic Alert Trigger: The system must have the ability to create an alert to any life-event record, which is automatically activated if certain conditions are met. Interface: The system must provide a Graphical User Interface (GUI) for data entry and associated activities. The interface must be user friendly, clear, predictable, consistent and efficient. Database Security: Due to the sensitive nature of the data elements that are intended to be held in the database, the proposed system must provide an encryption mechanism for securing the entire database (data at rest) as well as ensuring that transmitted data between two end points (externally and internally) are not compromised. As encryption affects data size and impacts on database performance, the system architecture should be designed to minimize any performance degradation. Audit Trail and Log: The system must provide a secure and automated audit capability for	

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Submission of Final Report & System Maintenance and Technical Support Submit Final Report and Implement a security architecture to protect data integrity and user information and ensure operational and maintenance (O&M), provide ongoing capacity support to ensure smooth operation of the platform, and offer technical support and maintenance services for 6 months post deployment.	

Task	Product/Deliverables	Schedule of Payments
Submission and acceptance of inception report.	• Inception Report	
System Design: The proposed system must include robust authentication and authorization controls, ensuring access only to registered users based on role-based privileges. It should enforce password strength and expiration policies, automatically log out users after failed login attempts, and record all access activities. Comprehensive database management tools should support user administration, access control, event scheduling, reporting, dynamic backup and recovery, and security management. The system must include an intuitive Graphical User Interface (GUI) to ensure efficient, consistent, and user-friendly data entry and navigation. Strong database security and encryption must protect both stored and transmitted data, with architecture optimized to maintain performance. An audit trail and log system must automatically record user activities, access attempts, and data changes, while preventing unauthorized modification or deletion of log files. The system should provide full reporting and printing capabilities, including predefined and ad-hoc reports, individualized profiles, and automated calculations. It must support data validation and formatting, automatically correcting input errors, enforcing mandatory fields, and providing warnings for invalid entries. Efficient search and retrieval functions must allow single or multi-criteria,	• Demo of Traditional Marriage Certificate Software • Submission of a Fully Automated Traditional Marriage Certificate Software	

wildcard, and case-insensitive searches, with duplicate detection and sorting features. Invalid or fraudulent records should be voided and archived (not deleted), accessible only to authorized users. Finally, the system must support data entry efficiency and keystroke reduction, including auto-population, drop-down menus, and dynamic suppression of unnecessary fields to streamline operations and ensure data integrity. Future Technological Integration: The system should be able to interface with other local automated systems. The system must have the ability to securely interconnect with these systems via a web-based interface. The web-based interface should support the use of Secure Socket Layer (SSL) for data transmission. The proposed system must ensure that incoming data input are validated, evaluated for expected size, format and type before acceptance. Because of the sensitive nature of the information being transmitted, the system must have the ability to reject unencrypted data.		
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Submission of Final Report & System Maintenance and Technical Support Submit Final Report and Implement a security architecture to protect data integrity and user information and ensure operational and maintenance (O&M), provide ongoing capacity support to ensure smooth operation of the platform, and offer technical support and maintenance services for 6 months post deployment.	• Submission of Final Report inclusive of an ongoing technical support plan.	

VI. Impact of Results
The development and deployment of the Traditional Marriage Certificate Software will have significant institutional, social, and governance impacts by modernizing and standardizing the registration and certification of traditional marriages to ensure accuracy, transparency, and security in record management. It will strengthen the Ministry of Internal Affairs' (MIA) capacity to manage civil registration processes digitally, reduce paperwork, enhance coordination between local and central authorities, and support data-driven decision-making through reliable marriage statistics. At the community level, the system will simplify access to legally recognized marriage certificates, promote legal rights, social protection, and gender equality, and prevent duplication, fraud, and record loss. Overall, it will foster citizens' trust in local governance, promote digital transformation, and advance decentralization by delivering efficient, transparent, and citizen-centered services.

VII. Evaluation Methodology and Criteria
The award of the contract is made to the individual consultant whose tender has been evaluated and found to be a) compliant/acceptable; and (b) Having received the highest score of weighted technical and financial criteria (Technical Criteria weight; 70%, Financial Criteria weight; 30%).

VIII. Technical Evaluation Criteria	
Summary	**Maximum total points**
Part 1: Relevance of education.	
Part 2: Relevance of experience in the area of specialization.	
Part 3: Other skills.	
Total	

No	Criteria	Maximum number of points
1.1	Relevance of Academic Qualifications: A minimum of a BBA or master's degree in computer science, software engineering, information systems, or a related field. Relevant professional certifications in software development, database management, cybersecurity, project management, or other similar qualification will be an added advantage.	
2.1	Relevance of experience in the area of specialization: A minimumseven years of professional experience in the design, development, and deployment of secure database management systems or civil registration and vital statistics (CRVS) solutions. Demonstrated expertise in developing web-based and mobile applications with strong authentication, encryption, and user management features. .	
2.2	A minimum of 7 years of proficiency in programming languages and frameworks such as Java, .NET, Python, PHP, or equivalent. Strong knowledge of database technologies (e.g., MySQL, PostgreSQL, MS SQL Server, Oracle) and data integration tools. Experience implementing Role-Based Access Control (RBAC), data validation, backup and recovery systems, and audit trail functionalities. Familiarity with cloud-based or hybrid hosting environments and data protection best practices.	
2.3	Prior experience developing digital solutions for government institutions, preferably in civil registration, local governance, or public administration systems. Familiarity with decentralization processes, legal frameworks, and traditional governance structures in Liberia or similar contexts. Demonstrated ability to design systems that ensure interoperability with existing national databases and information systems. Strong analytical skills for system design, business process mapping, and requirements analysis. Proven ability to produce high-quality documentation including	

	technical specifications, user manuals, and system maintenance guides. Ability to develop user-friendly interfaces suitable for both central and local-level users.	
3.1	Proven experience working with United Nations agencies and international development organizations.	
3.2	Demonstrated levels of proficiency in English	